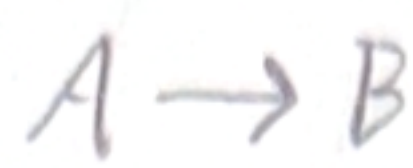


CHE-312

11/17/25

HW #9

Q1

 $C_A(y=L) = \text{impermeable, unreactive}$

$C_A(y=0) = C_0$

(0th order rxn) α and D_{A0}
constant

$$D_A = \frac{D_{A0}}{1 + \alpha y}, \quad R_{AB} = -K_B, \quad R_{AF} = -K_F$$

$$\frac{d}{dy} \left(D_A \frac{dC_A}{dy} \right) - K_B - K_F = 0 \quad (\text{Balance Eq})$$

$$\text{Total rate of reactant A consumed} = N_A'' = -D_A \frac{dC_A}{dy}$$

$$\Rightarrow D_A \frac{dC_A}{dy} = (K_B + K_F)y + A$$

$$\Rightarrow \frac{dC_A}{dy} = \frac{(K_B + K_F)y + A}{D_{A0}} \Rightarrow \frac{(1 + \alpha y)(y(K_B + K_F) + A)}{D_{A0}}$$

$$\Rightarrow \frac{dC_A}{dy} = \frac{y(K_B + K_F) + A + \alpha y^2(K_B + K_F) + A\alpha y}{D_{A0}}$$

$$\Rightarrow C_A(y) = \frac{y^2(K_B + K_F)}{2D_{A0}} + \frac{Ay}{D_{A0}} + \frac{\alpha y^3(K_B + K_F)}{3D_{A0}} + \frac{A\alpha y^2}{2D_{A0}} + B$$

$$\Rightarrow C_A(y) = \frac{y^3 \alpha (K_B + K_F)}{3D_{A0}} + \frac{y^2((K_B + K_F) + A\alpha)}{2D_{A0}} + \frac{Ay}{D_{A0}} + B$$

$$\text{BC-1)} \quad C_A(y=0) = C_0 \Rightarrow \boxed{B = C_0}$$

$$\text{BC-2)} \quad \left. \frac{dC_A}{dy} \right|_{y=L} = 0$$

$$\frac{d}{dy} (C_A(y)) = \frac{\cancel{3}y^2 \alpha (K_B + K_F)}{\cancel{3}D_{A0}} + \frac{\cancel{2}y(K_B + K_F)}{\cancel{2}D_{A0}} + \frac{\cancel{2}yA\alpha}{\cancel{2}D_{A0}} + \frac{A}{D_{A0}}$$

$$\left. \frac{dC_A}{dy} \right|_{y=L} = \frac{L^2 \alpha (K_B + K_F)}{D_{A0}} + \frac{L(K_B + K_F)}{D_{A0}} + \frac{L(A\alpha)}{D_{A0}} + \frac{A}{D_{A0}} = 0$$

$$\frac{AL\alpha}{D_{A0}} + \frac{A}{D_{A0}} = - \frac{L(K_B + K_F)}{D_{A0}} (L\alpha + 1)$$

$$\Rightarrow A(L\alpha + 1) = -L(K_B + K_F)(L\alpha + 1)$$

$$\Rightarrow \boxed{A = -L(K_B + K_F)}$$

$$C_A(y) = \frac{y^3 \alpha (K_B + K_F)}{3 D_{A0}} + \frac{y^2 ((K_B + K_F) - L(K_B + K_F)\alpha)}{2 D_{A0}} - \frac{L(K_B + K_F)y}{D_{A0}} + C_0$$

$$\Rightarrow \boxed{\frac{y^3 \alpha (K_B + K_F)}{3 D_{A0}} + \frac{y^2 (K_B + K_F)(1 - L\alpha)}{2 D_{A0}} - \frac{y L (K_B + K_F)}{D_{A0}} + C_0}$$

$$(N_A'' = D_A \frac{dC_A}{dy})$$

$$\frac{-D_{A0}}{1 + \alpha y} \left(\frac{y^2 \alpha (K_B + K_F)}{D_{A0}} + \frac{y (K_B + K_F)(1 - L\alpha)}{D_{A0}} - \frac{L (K_B + K_F)}{D_{A0}} \right)$$

$$\Rightarrow \boxed{N_A'' = \frac{(K_B + K_F)(y^2 \alpha + y(1 - L\alpha) - L)}{1 + \alpha y}}$$

Since reactants are consumed at every point at $y=0$ or further along (up until L), the net flux will be $N_A''|_{y=0}$

$$\Rightarrow N_A''|_{y=0} = \frac{(K_B + K_F)((0)^2 \alpha + (0)(1 - L\alpha) - L)}{1 + \alpha(0)}$$

* missed negative

$$\Rightarrow N_A''|_{y=0} = -L(K_B + K_F) = \boxed{L(K_B + K_F)}$$

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HW #9

Q_2 $\dot{m}_H \cdot C_{p,H} = 600 \frac{W}{K}$, $\dot{m}_C \cdot C_{p,C} = 800 \frac{W}{K}$

$U = 400 \frac{W}{m^2 \cdot K}$ $T_{H,i} = 110^\circ C$ $T_{H,o} = 80^\circ C$

$T_{C,i} = 30^\circ C$ $T_{C,o} = ?$

OF to K

OF

255.928K

$\dot{q} = \dot{m}_H C_{p,H} (T_{H,i} - T_{H,o}) = \dot{m}_C C_{p,C} (T_{C,o} - T_{C,i})$

$\Rightarrow T_{C,o} = \frac{\dot{m}_H C_{p,H} (T_{H,i} - T_{H,o})}{\dot{m}_C C_{p,C}} + T_{C,i}$

$\Rightarrow \frac{(600 \frac{W}{K})((110^\circ C - 80^\circ C) + 273)K}{800 \frac{W}{K}} + ((30^\circ C + 273)K)$

$\Rightarrow T_{C,o} = 325.5 K$ $T_{H,i} = 383 K$ $T_{H,o} = 353 K$

$T_{C,i} = 303 K$

$\dot{q} = U A \frac{(\Delta T_2 - \Delta T_1)}{\ln(\Delta T_2 / \Delta T_1)}$

(Counter-current) $\Delta T_1 = T_{H,i} - T_{C,o}$, $\Delta T_2 = T_{H,o} - T_{C,i}$

(Co-current) $\Delta T_1 = T_{H,i} - T_{C,i}$, $\Delta T_2 = T_{H,o} - T_{C,o}$

$\dot{q} = (600 \frac{W}{K})((110 + 273) - (80 + 273)) = 18,000 W$

i) $A = \frac{\dot{q} (\ln(\Delta T_2 / \Delta T_1))}{U (\Delta T_2 - \Delta T_1)}$ $\Delta T_1 = 57.5 K$ $\Delta T_2 = 50 K$

$\frac{(18000 W) \ln(50/57.5)}{(400 \frac{W}{m^2 \cdot K})(50 K - 57.5 K)} = A_{counter} = 0.839 m^2$

ii) $\Delta T_1 = 80 K$, $\Delta T_2 = 27.5$

$\frac{(18000 W) (\ln(\frac{27.5}{80}))}{(400 \frac{W}{m^2 \cdot K})(27.5 K - 80 K)} = A_{co} = 0.915 m^2$